MAT 2305- Real Analysis II
Assignment 03

Answers all the questions and submit the **solutions** for **1(ii), 2(i), 4, 8,10, 11, 13, 15** on or before **17th July, 2024**

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1. Use the Chain Rule to find dz/dt or dw/dt.
   i. $z = x^2 y + xy^2, x = 2 + t^4, y = 1 - t^3$
   ii. $w = \ln\sqrt{x^2 + y^2 + z^2}, x = \sin t, y = \cos t, z = \tan t$.
   iii. $z = \sqrt{x^2 + y^2}, x = e^{2t}, y = e^{-2t}$.
2. Use the Chain Rule to find $\partial z/\partial s$ and $\partial z/\partial t$.
   i. $z = x^2 y^3, x = s\cos t, y = s\sin t$
   ii. $z = e^{x+2y}, x = s/t, y = t/s$
   iii. $z = \tan(u/v), u = 2s + 3t, v = 3s - 2t$
3. Show that the Cobb –Douglas production function $P = bL^{\alpha} K^{\beta}$ satisfies the equation
$$L\frac{\partial P}{\partial L} + K\frac{\partial P}{\partial K} = (\alpha + \beta)P.$$
4. if $z = e^{ax+by} f(ax - by)$, prove that
$$b\frac{\partial z}{\partial x} + a\frac{\partial z}{\partial y} = 2abz,$$
   where a and b are constants.
5. If $v = (x^2 + y^2 + z^2)^{m/2}$, the determine the value of $m(\neq 0)$ which will make
$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = 0.$$
6. If $z = f(x, y)$, where $x = s + t$ and $= s - t$.show that
$$\left(\frac{\partial z}{\partial x}\right)^2 - \left(\frac{\partial z}{\partial y}\right)^2 = \frac{\partial z}{\partial s}\frac{\partial z}{\partial t}.$$
7. If $u = f(x, y)$, where $x = e^s \cos t$ and $y = e^s \sin t$, show that
$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = e^{-2s}\left(\frac{\partial^2 u}{\partial s^2} + \frac{\partial^2 u}{\partial t^2}\right).$$

8. Find the local maximum and minimum values and the saddle point(s) of the function f :

   i. $f(x,y) = 9 - 2x + 4y - x^2 - 4y^2$
   ii. $f(x,y) = xy - 2x - y$
   iii. $f(x,y) = 2x^3 + xy^2 + 5x^2 + y^2$
   iv. $f(x,y) = e^x \cos y$.

9. Find the absolute maximum and minimum values of $f(x,y) = x^4 + y^4 - 4xy + 2$ on the set $D = \{(x,y) \mid 0 \le x \le 3, 0 \le y \le 2\}$.
10. Find the dimensions of the rectangle box with largest volume if the total surface area is given as 64 cm^2.
11. Find the points on the surface that are $y^2 = 9 + xz$ that are closest to the origin.
12. Find the points on the surface that are $z^2 = x^2 + y^2$ that are closest to the point $(4,0.2)$.
13. Use Lagrange multipliers to find the maximum and minimum values of the function f subject to the given constraint(s):

   i. $f(x,y) = x^2 + y^2 : xy = 1$.
   ii. $f(x,y,z) = x^2 + y^2 + z^2; x^4 + y^4 + z^4 = 1$
   iii. $f(x,y,z) = yz + xy; xy = 1, y^2 + z^2 = 1$
   iv. $f(x,y,z) = 3x - y - 3z; x + y - z = 0, x^2 + 2z^2 = 1.$

14. Use Lagrange multipliers to prove that the rectangle with maximum area that has a given perimeter p is a square.
15. Find the dimensions of rectangular box with largest volume if the total surface area is given as $64\,cm^2$.
16. Find the volume of the largest rectangular box in the first octant with three faces in the coordinate planes and one vertex in the plane $x + 2y + 3z = 6$.
17. Fins three positive numbers whose sum is 12 and the sum of whose squares is as small as possible.
18. The base of an aquarium with given volume V is made of slate and sides are made of glass. If slate costs five times as much (per unit area) as glass, find the dimensions of the aquarium that minimize the cost of materials.